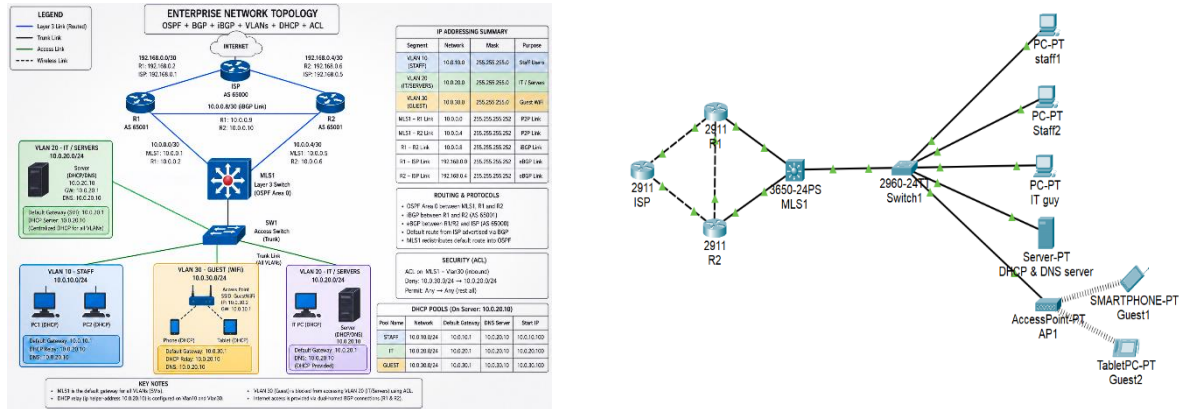


Enterprise Network Lab Documentation

Enterprise Network Design and Implementation Using OSPF, BGP, and VLAN Segmentation



1. Introduction

Modern enterprise networks require scalability, security, redundancy, and efficient resource management. This project presents the design and implementation of a simulated enterprise network using Cisco Packet Tracer. The objective was to build a structured, multi-layered network that reflects real-world networking practices, incorporating dynamic routing protocols, segmentation techniques, centralized services, and controlled access policies.

The network was designed to support multiple departments, wireless guest access, and dual-homed internet connectivity. Emphasis was placed not only on functionality but also on the reasoning behind each design decision, ensuring alignment with industry standards and best practices.

2. Problem Statement and Objectives

The primary goal of this project was to design a network that satisfies the following requirements:

- Provide logical separation between different user groups
- Enable communication between internal networks while maintaining security boundaries
- Ensure redundancy in external connectivity
- Support centralized services such as DHCP and DNS
- Allow controlled wireless access for guest users
- Implement security policies to restrict unauthorized access
- Demonstrate the ability to troubleshoot and resolve network failures

The project also aimed to simulate real-world enterprise conditions by introducing routing complexity and enforcing structured troubleshooting methodologies.

3. Network Topology and Architecture

The network follows a hierarchical design consisting of core, distribution, and access layers.

At the core, a Layer 3 switch (MLS1) performs inter-VLAN routing and acts as the central point for internal traffic flow. Two edge routers (R1 and R2) connect the internal network to an upstream ISP, providing redundant external connectivity. An access layer switch (SW1) connects end-user devices and the wireless access point.

The topology includes:

- A core Layer 3 switch responsible for routing between VLANs
- Two edge routers providing redundancy and external connectivity
- A simulated ISP router to emulate internet access
- A server hosting DHCP and DNS services
- A wireless access point serving guest devices
- End-user devices distributed across VLANs

This structure ensures separation of concerns, scalability, and fault tolerance.

4. IP Addressing Scheme and Design Logic

A structured IP addressing scheme was implemented to ensure clarity, scalability, and efficient use of address space.

Three primary VLANs were defined:

- VLAN 10 (Staff): 10.0.10.0/24
- VLAN 20 (IT): 10.0.20.0/24
- VLAN 30 (Guest): 10.0.30.0/24

Each VLAN represents a distinct functional group within the organization. The first usable IP address in each subnet was reserved as the default gateway, implemented on the Layer 3 switch. This convention simplifies network management and aligns with industry practices.

Point-to-point links between routers and switches were assigned /30 subnets. This decision minimizes address waste while providing sufficient addressing for two endpoints per link. For example, links between the core switch and edge routers, as well as inter-router connections, utilize /30 networks.

The IP addressing scheme was designed to be predictable and extensible, allowing for future expansion without major redesign.

5. VLAN Segmentation and Network Organization

VLANs were implemented to logically segment the network into distinct broadcast domains. This segmentation improves security, reduces broadcast traffic, and enhances overall network performance.

The Staff VLAN accommodates employee devices, the IT VLAN hosts critical infrastructure such as servers, and the Guest VLAN provides isolated access for wireless users. By separating these groups, the network ensures that sensitive resources are not directly exposed to less trusted users.

Inter-VLAN routing is performed by the Layer 3 switch, enabling controlled communication between VLANs while maintaining logical separation.

6. Routing Design and Protocol Selection

6.1. OSPF (Open Shortest Path First)

OSPF was selected as the internal routing protocol due to its scalability, fast convergence, and widespread use in enterprise environments. It operates as a link-state protocol, allowing routers to maintain a consistent view of the network topology.

OSPF was configured between the Layer 3 switch and both edge routers. This enables dynamic route exchange for all internal networks, eliminating the need for static routing and simplifying network management.

6.2. BGP (Border Gateway Protocol)

BGP was implemented on the edge routers to simulate communication with an external ISP. Each router establishes an external BGP (eBGP) session with the ISP, enabling the exchange of routing information.

BGP was chosen because it is the standard protocol for inter-domain routing and provides flexibility in controlling traffic flow through policy-based decisions.

6.3. Internal BGP (iBGP)

An internal BGP session was configured between the two edge routers. This allows them to share routes learned from the ISP, ensuring redundancy and consistent routing decisions across the network.

The inclusion of iBGP ensures that if one router loses connectivity to the ISP, the other can still provide a valid path to external networks.

6.4. Default Route Propagation

The ISP advertises a default route to the edge routers. This route is then propagated into the internal network via OSPF using the default-information originate command. This allows all internal devices to reach external networks without requiring explicit route definitions.

7. Services Implementation

7.1. DHCP and DNS

A centralized server located in the IT VLAN provides both DHCP and DNS services. This design centralizes network services, simplifying management and improving consistency. DHCP pools were created for each VLAN, assigning IP addresses dynamically to clients. DNS services allow devices to resolve hostnames, enhancing usability.

7.2. DHCP Relay

Because the DHCP server resides in a different VLAN from some clients, DHCP relay was configured on the Layer 3 switch. This allows DHCP requests from other VLANs to be forwarded to the server.

Without DHCP relay, devices in separate VLANs would be unable to obtain IP addresses automatically.

8. Wireless Network Design

A wireless access point was deployed to provide connectivity for guest devices. The access point is connected to the access switch and assigned to the Guest VLAN.

Wireless clients connect using a secured SSID configured with WPA2-PSK. This ensures that only authorized users can access the network while maintaining isolation from internal resources.

The design ensures that wireless traffic is treated as part of the Guest VLAN, subject to the same restrictions and policies.

9. Security Implementation

Security was enforced using extended Access Control Lists (ACLs). The primary policy implemented was to prevent guest users from accessing sensitive IT resources.

An ACL was applied inbound on the Guest VLAN interface to block traffic destined for the IT VLAN while allowing all other traffic. This ensures that guest users can access external resources without compromising internal systems.

The placement of the ACL at the Layer 3 switch ensures that unwanted traffic is blocked as early as possible, reducing unnecessary load on the network.

10. Challenges and Issues Encountered

Several issues were encountered during implementation, each providing valuable troubleshooting experience.

One significant issue involved the inability of guest devices to access external networks. Initially, this appeared to be an ACL misconfiguration. However, further investigation revealed that the Layer 3 switch lacked a default route. The issue was resolved by propagating the default route through OSPF.

Another issue involved DHCP failures across VLANs, which was traced to the absence of DHCP relay configuration. Once the helper address was configured, clients were able to obtain IP addresses successfully.

Additionally, misconfigured ACL rules initially blocked all traffic due to the absence of a permit statement. This highlighted the importance of rule order and completeness in ACL design.

11. Troubleshooting Methodology

A structured troubleshooting approach was consistently applied throughout the project. The process followed a logical progression:

Step-by-step approach:

1. Verify interface status and connectivity

`show ip interface brief`

2. Check routing tables for missing or incorrect routes

`show ip route`

3. Check OSPF neighbor relationships and route propagation

`show ip ospf neighbor`

4. Validate BGP session status and route exchange

`show ip bgp summary`

5. Inspect ACL configurations and hit counters

`show access-lists`

This systematic approach ensured efficient fault isolation and resolution, minimizing guesswork and reinforcing best practices.

12. Design Justification

Each design decision was made with specific objectives in mind. VLAN segmentation was used to enhance security and organization. OSPF was selected for its efficiency in internal routing, while BGP provided realistic external connectivity and policy control.

The use of dual edge routers introduced redundancy, ensuring high availability. Centralized services improved manageability, and ACLs enforced security policies at the network boundary.

The overall design reflects a balance between simplicity and realism, making it suitable as both a learning exercise and a demonstration of practical networking skills.

13. Conclusion

This project successfully demonstrates the design and implementation of a structured enterprise network incorporating modern networking principles. It integrates multiple technologies, including VLANs, dynamic routing protocols, centralized services, and security mechanisms.

Beyond implementation, the project emphasizes understanding, troubleshooting, and documentation. By simulating real-world challenges and resolving them systematically, it provides a strong foundation for practical network engineering.

The resulting network is scalable, secure, and resilient, reflecting the core requirements of modern enterprise environments.